

Smart Wall Paint Visualizer Using MEAN Stack

Abstract

The Smart Wall Paint Visualizer is an innovative web-based application developed using the MEAN Stack, which consists of MongoDB, Express.js, Angular, and Node.js. The primary objective of this system is to help users visualize different wall paint colors and designs on their room images before actual painting. Selecting the right wall color is often difficult because traditional paint catalogs and sample cards do not accurately represent how a color will appear under real lighting conditions and room dimensions. The proposed system provides a digital solution by allowing users to upload room photographs and apply virtual paint colors to specific wall areas. Through image processing and AI-based wall detection techniques, users can preview realistic painting results, reducing uncertainty and improving decision-making. This application serves homeowners, interior designers, architects, and paint manufacturers by providing an efficient and cost-effective visualization platform.

Introduction

Color selection plays a vital role in interior design and home decoration. The appearance of a room can change significantly depending on the wall colors used. Traditionally, customers rely on paint sample books, brochures, or showroom displays when choosing paint colors. However, these methods often fail to provide a realistic understanding of how a color will look in a specific room environment. Factors such as lighting, room size, furniture arrangement, and wall texture influence the final appearance of paint.

The Smart Wall Paint Visualizer addresses this challenge by enabling users to upload photographs of their rooms and digitally apply different paint colors. The application provides a real-time preview, helping users make informed decisions before investing in actual painting. This reduces the risk of dissatisfaction, saves time, and minimizes expenses associated with repainting.

The project is developed using the MEAN Stack, which provides a robust, scalable, and efficient architecture for modern web applications. MongoDB manages project data, Express.js and Node.js handle backend operations, and Angular delivers a responsive user interface.

Objectives

The main objectives of the Smart Wall Paint Visualizer are:

1. To provide a realistic virtual painting experience.
2. To allow users to upload room images and preview paint colors.
3. To reduce uncertainty in paint color selection.
4. To save customized paint projects for future reference.

5. To implement image processing techniques for wall detection.
6. To create a user-friendly and responsive web application.

Methodology

The system follows a full-stack architecture. Users first register and log in to the application. After authentication, they can upload a room image through the frontend interface. The uploaded image is processed using image analysis techniques such as OpenCV.js or TensorFlow.js to identify wall regions.

Once the wall area is detected, the user can select colors from a predefined palette or custom color picker. The selected color is then applied to the wall region while preserving shadows and textures to create a realistic appearance. The generated preview is displayed instantly on the screen.

The backend server processes requests using Node.js and Express.js, while MongoDB stores user information, uploaded images, and saved projects. The system also provides options to save and download the final painted image.

System Architecture

The application architecture consists of four major components:

Frontend (Angular):

- User Interface
- Image Upload Module
- Color Selection Panel
- Preview Dashboard

Backend (Node.js & Express.js):

- Authentication Services
- Image Processing APIs
- Project Management APIs

Database (MongoDB):

- User Data Storage
- Project Storage
- Image Metadata Storage

AI/Image Processing Layer:

- **Wall Detection**
- **Mask Generation**
- **Color Overlay Application**

Results and Discussion

The Smart Wall Paint Visualizer successfully demonstrates how modern web technologies can be integrated with image processing techniques to solve real-world interior design problems. Users can visualize multiple paint colors within seconds, making the color selection process more efficient and reliable.

The system improves customer satisfaction by providing realistic previews before painting. It also benefits paint companies by offering a digital marketing and visualization platform. Preliminary testing shows that the application significantly reduces the likelihood of incorrect color selection and enhances user confidence.

Conclusion

The Smart Wall Paint Visualizer is an effective and practical solution for virtual interior painting. By combining the capabilities of the MEAN Stack with AI-powered image processing, the application enables users to preview wall colors realistically and make informed decisions. The system is scalable, user-friendly, and suitable for both residential and commercial use. Future enhancements may include AI-based color recommendations, texture simulation, augmented reality integration, and 3D room visualization to further improve the user experience.

References

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